



* Lab 1 : NMAP : Network Mapping and Port Scanning

Objective: To understand how to use NMAP for TCP port ~~sc~~ scanning and perform network monitoring tasks.

Tools Required: NMAP

Theory : NMAP is a powerful open-source network scanning tool used to discover hosts, services, and open ports on a network. It supports a wide range of scanning options like SYN scan, version detection, OS detection, and more.

Procedure:

1. Basic Host Discovery:

```
nmap -sn 192.168.1.0/24
```

2. TCP Port Scan:

```
nmap -sT 192.168.1.1
```

3. SYN Scan (Stealth Scan):

```
nmap -sS 192.168.1.1
```



4. OS Detection :

```
nmap -O 192.168.1.1
```

5. Service Version Detection :

```
nmap -sV 192.168.1.1
```

6. Scan Multiple IPs :

```
nmap 192.168.1.1 192.168.1.2
```

7. Scan IP Range

```
nmap 192.168.1.1-20
```

8. Aggressive Scan :

```
nmap -A 192.168.1.1
```

9. Detect Firewalls :

```
nmap -PN 192.168.1.1
```

10. Output Scan to File :

```
nmap -oN result.txt 192.168.1.1
```



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** Output **